

Strategy Brief #8

Monitoring and Modelling Future Land Subsidence



Exploring how satellite-based remote sensing (InSAR) and predictive modelling can close the critical gap in subsidence monitoring across developing coastal regions, and why integrating these tools into urban and infrastructure planning is essential to shifting from costly reactive responses to evidence-based proactive risk management.

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Introduction

Land subsidence is increasingly recognized as a critical risk in coastal and urban regions, particularly in developing countries where **monitoring systems are limited**. Advances in remote sensing and modelling now allow for improved detection and forecasting of subsidence patterns.



Key Challenge

Most regions lack **integrated monitoring and predictive systems**, limiting the ability to anticipate subsidence and incorporate it into infrastructure and urban planning decisions.

Strategic Insights

- Satellite-based techniques (e.g., InSAR) enable **cost-effective large-scale monitoring**.
- Subsidence is often driven by **human activities (e.g., groundwater extraction, urban load)**.
- Predictive modelling can identify **future high-risk zones**.
- Current planning frameworks rarely integrate **subsidence projections**.



Strategic Implications

- Infrastructure and urban investments are exposed to **unanticipated long-term risks**.
- Lack of monitoring leads to **reactive rather than proactive responses**.
- Increasing financial and environmental costs over time.



Recommendations

- Establish **national and regional subsidence monitoring systems using remote sensing.**
- Develop **predictive models integrating climate, land use, and human activities.**
- Integrate subsidence data into **urban planning and infrastructure design.**
- Strengthen collaboration between **research institutions and policymakers.**

Value Proposition

Proactive monitoring and modelling enable **early risk detection, better planning, and significant cost savings** in infrastructure and urban development.